Finite kernels of full-rank orthogonal inverters in characteristic two

Human Compression project

Let q be even, let $m \geq 2$, let V be a nonsingular quadratic space of dimension $2m$, and let $t \in O(V)$ have $(t-I)^2 = 0$ and rank m. Write B for the polar form and assume $B(v, tv)$ is not identically zero. These are the full-rank types c_m for even m, and b_m for odd m in Aschbacher–Seitz Sections7–8. The result below also covers $t \notin \Omega(V)$: the centralizer is always taken in $\Omega(V)$. Put $\nu_2(H) = |\{h : h^2 = 1\}|$.

Theorem 1. *There are surjective homomorphisms with the indicated finite kernel orders:*

	image of $C_{\Omega(V)}(t)$	*kernel order*
m odd	$\mathrm{Sp}_{m-1}(q)$	$q^{(m^2-m)/2}$
m even	$\mathrm{Sp}_{m-2}(q)$	$q^{(m^2+m-2)/2}$.

Here Sp_0 *is trivial. Consequently the centralizer's* ν_2 *is at most the kernel order times* ν_2 *of the displayed symplectic group. In particular, when* $m = 4$*,*

$$\nu_2(C_{\Omega(V)}(t)) \leq q^9 q^2 = q^{11}.$$

No order is inferred from an algebraic dimension.

1 A direct finite fibre calculation

The nonalternating full-rank symplectic normal form from AS7.6–7.7 permits coordinates $x, y \in \mathbb{F}_q^m$ in which

$$t(x, y) = (x + y, y), \qquad B((x, y), (x', y')) = x^t y' + y^t x'.$$

Equivalently, the nonsingular nonalternating symmetric matrix defining the square-zero part is congruent to I_m over the perfect field. Let $s = (1, \ldots, 1)^t$. A quadratic form with this polar form differs from $x^t y$ by squares of linear forms. Its invariance under t forces its x-part to be $(s^t x)^2$. Thus, for some column b, it is exactly

$$Q(x, y) = (s^t x)^2 + x^t y + b^t y^{[2]}, \tag{1}$$

where [2] denotes entrywise squaring.

Block commutation and the symplectic equation give every symplectic centralizer element uniquely as

$$g_{A,D}(x, y) = (Ax + DAy, Ay), \qquad AA^t = I_m, \quad D = D^t.$$

Put $J_m = \{A : AA^t = I_m\}$. These matrices fix s, since $v^t v = (s^t v)^2$ and squaring is injective. Substituting $w = Ay$ in (1) proves that $g_{A,D}$ is orthogonal precisely when

$$F(D) := \mathrm{diag}(D) + (Ds)^{[2]} = b + A^{[2]} b. \tag{2}$$

This is an equation of additive groups, not an asserted $\mathbb{F}_q$-linear map.

We count its fibres. Put $d = m(m-1)/2$. If c_i is the sum of the off-diagonal entries in row i, then $\sum c_i = 0$. The map from the d off-diagonal entries to (c_i) is onto that hyperplane: entries in the last column prescribe its first $m-1$ coordinates. Every fibre has q^{d-m+1} elements. Writing $a_i = D_{ii}$, equation $F(D) = 0$ becomes

$$a_i^2 + a_i = c_i^2.$$

Over $\mathbb{F}_q$, the map $a \mapsto a^2 + a$ has kernel of order two and image $\ker \mathrm{Tr}_{q/2}$. Since $\mathrm{Tr}(c_i^2) = \mathrm{Tr}(c_i)$, there are $(q/2)^{m-1}$ possible row-sum vectors, then q^{d-m+1} off-diagonal choices, and 2^m diagonal choices. Therefore

$$|\ker F| = 2q^d. \tag{3}$$

Also, writing $a = \sum_i D_{ii}$, we have $\sum_i F(D)_i = a + a^2$. It follows that the image of F lies in $\{c : \mathrm{Tr}_{q/2}(\sum c_i) = 0\}$. Both sets have size $q^m/2$ by (3), so they are equal. Since $s^t A^{[2]} = s^t$, the right side of (2) has coordinate sum zero. Thus every $A \in J_m$ lifts, and every fibre in $C_{O(V)}(t) \to J_m$ has exactly $2q^d$ elements.

The matrix $g_{I,E_{11}}$ is in this kernel: $F(E_{11}) = 0$. It has rank one after subtraction of the identity, so its Dickson invariant is one. We retain the standard Dickson homomorphism $O(V) \to C_2$, $g \mapsto \mathrm{rank}(g - I) \bmod 2$, whose kernel is $\Omega(V)$. Multiplication by this kernel element toggles that invariant without changing A. Therefore every fibre has exactly q^d elements in $\Omega(V)$. We have proved the surjection

$$C_{\Omega(V)}(t) \longrightarrow J_m, \qquad |\ker| = q^d. \tag{4}$$

This argument pays the full orthogonal component and its index-two intersection; no connectedness or splitting assumption replaces the count.

2 The smaller symplectic image

For odd m, $s^t s = 1$, and $s^\perp$ is a nonsingular alternating space. Every member of J_m fixes s, and conversely every symplectic isometry of $s^\perp$ extends by fixing s. Hence $J_m = \mathrm{Sp}_{m-1}(q)$.

For even m, $s^t s = 0$. Choose e with $s^t e = 1$, whence $e^t e = 1$, and let $H = \langle s, e\rangle^\perp$. The form on H is nonsingular alternating of dimension $m-2$. The action on $s^\perp/\langle s\rangle$ gives a surjection $J_m \to \mathrm{Sp}(H)$, split by fixing s, e. Every element of its kernel, and no other one, has the form

$$s \mapsto s, \qquad h \mapsto h + (h^t u)s \ (h \in H), \qquad e \mapsto e + u + as, \qquad u \in H,\ a \in \mathbb{F}_q.$$

These maps preserve every displayed pairing. Conversely, fixing the quotient forces the stated shape, and the pairing with e determines its linear functional. Thus the kernel has exactly q^{m-1} elements. Combining with (4) proves the exact kernel orders in Theorem 1. These kernels have orders which are powers of two. They are therefore unipotent matrix groups in characteristic two; their orders above are finite counts, including the orthogonal component factor.

For any surjection $C \to L$ with finite kernel K, an element of square one maps to an element of square one. There are at most $|K|$ such elements above each image, so $\nu_2(C) \le |K|\nu_2(L)$. In characteristic two, $\mathrm{Sp}_2(q) = \mathrm{SL}_2(q)$ has exactly $q^2 - 1$ nonidentity involutions: each is a rank-one unipotent, with one eigenline, and for each of the $q+1$ lines there are $q-1$ nontrivial transvections. Thus $\nu_2(\mathrm{Sp}_2(q)) = q^2$, which proves the stated q^{11} bound.

3 Application to the natural torus inverters

In the acquired orthogonal torus models, a natural projective involutory inverter in characteristic two has an isometric lift of square one. Indeed if a conformal lift has square aI, its multiplier is a;

rescaling by the unique square root of a^{-1} achieves both conditions. Projective inversion becomes literal inversion, since the only scalar isometry in characteristic two is one.

On a negative $2k$-block write $E = \mathbb{F}_{q^{2k}}$, $\bar{v} = v^{q^k}$ and $Q(v) = \mathrm{Tr}_{q^k/q}(c_0 v\bar{v})$, $c_0 \neq 0$. The acquired torus degree argument distinguishes this irreducible block from the others. If its multiplier is α, then $\alpha^{-1} = \bar{\alpha}$, and an inverting linear map satisfies $t(\alpha v) = \bar{\alpha} t(v)$. Since $\mathbb{F}_q[\alpha] = E$, it is exactly $v \mapsto c\bar{v}$. Its square is one precisely when $c\bar{c} = 1$. The small negative plane in the plus model has the same argument, since its primitive multiplier has order $q + 1 > 2$. Hilbert90, or the cyclic multiplicative group directly, conjugates it linearly to $v \mapsto \bar{v}$, so its square-zero part has rank k. Moreover its polar form satisfies

$$B(v, c\bar{v}) = \mathrm{Tr}_{E/\mathbb{F}_q}(c_0 \bar{c}\, v^2),$$

which is not identically zero by the nondegenerate trace pairing and bijectivity of squaring. The minus model therefore has rank m and is nonalternating. In the plus model the two negative blocks give ranks $m - 1$ and one; restriction to the large block already makes the full form nonalternating. They therefore have precisely the types c_m (m even) and b_m (m odd), and Theorem 1 applies for either orthogonal sign. No flag-dimension test is required to eliminate a_m.

Source and scope. AS1976 Sections7–8 supplies the symplectic normal form and class labels; the complete relevant passages and the 1978 corrections were read. The correction to AS8.8(i) concerns the last column of P; here the middle space has dimension zero, and the direct equation (2) independently pays all finite fibres. The Dickson convention is retained from Murray–Roney-Dougal2008, Definition3.1(2), Lemma3.2 and Definition3.3. The coordinate fibre count and its use here are new derivations by publication_recovery; root requested this interface and proposed its expected exponents before this proof. These arguments do not pay triality, maximal-overgroup classification, odd-characteristic fixed groups or the whole L045 claim.

References

[1] M. Aschbacher and G. M. Seitz, *Involutions in Chevalley groups over fields of even order*, Nagoya Math. J. 63 (1976), 1–91, Sections 7–8; corrections, Nagoya Math. J. 72 (1978), 135–136.

[2] S. H. Murray and C. M. Roney-Dougal, *The spinor norm and homomorphism algorithms for classical groups*, preprint dated 2 May 2008, Section 3.